

Srinija J
Sr. Data Scientist AI/ML
+1 254-307-5203
srinijajannapureddy1@gmail.com
[linkedin.com/in/SrinijaJ](https://www.linkedin.com/in/SrinijaJ)

Professional Summary:

- 10 + years of Data Scientist with hands-on experience in data analysis and a track record of creating actionable insights for complex high- tech products. Professional expertise includes:
- establishing min data set of features and baselines.
- prototyping anomaly detection algorithm based on multiple subsets of features, performing dimensionality reduction.
- Performed data pre-processing and cleaning to prepare data sets for further statistical analysis; including outlier detection and treatment, missing value treatment, variable transformation and various other data manipulation techniques using SAS programming language.
- Involved in Analyzing the Business Requirements.
- Extensively used Datastage Designer to develop processes for extracting, transforming, integrating and loading data from various sources into the Data Warehouse database.
- Worked closely with Business Analyst to understand business requirements and start deriving solutions accordingly.
- Work with DevOps teams to ensure scalability, reliability, and security of AI solutions.
- Worked on natural language processing for documentation classification, text processing using NLTK, SPACY, TextBlob to find the sensitive information in the electronically stored files and text summarization.
- Worked on various machine learning algorithms like Linear regression, logistic regression, Decision trees, random forests, K- means clustering, Support vector machines, XGBoosting on client requirements.
- Developed machine learning models using recurrent neural networks – LSTM for time series, predictive analytics.
- Good knowledge on image classification problems using the Keras Models for image classification with weights trained on ImageNet like VGG16, VGG19, ResNet, ResNetV2, InceptionV3. Knowledge on OpenCV for real time computer vision.
- Data Preparation for raw data arriving in many different formats, Prototyping & Build-out of Analytical Models to be used in establishing most effective baseline for further computation.

Technical Skills:

Statistical Methods	Hypothesis Testing, ANOVA, Principal Component Analysis (PCA), Time Series, Correlation (Chi-square test, covariance), Multivariate Analysis, Bayes Law.
Deep Learning	Artificial Neural Networks, Convolutional Neural Networks, RNN, Deep Learning on AWS, Keras API.
Hadoop Ecosystem	Hadoop, Spark, MapReduce, Hive QL, HDFS, Sqoop, Pig
Data Visualization	Tableau, Python (Matplotlib, Seaborn), R(ggplot2), Power BI, QlikView, D3.js
Languages	Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn), R, SQL, MATLAB, Spark, Java, C#
Operating Systems	UNIX Shell Scripting (via PuTTY client), Linux, Windows, Mac OS

Project Experience:

Client: CVS Health. "Scottsdale, AZ

April 2023 - Present

Role: Senior Data Scientist

Responsibilities:

- Elucidating the continuous improvement opportunities of current predictive modeling algorithms. Proactively collaborates with business partners to determine identified population segments and develop actionable plans to enable the identification of patterns related to quality, use, cost and other variables.
- Developed the Python automation script for consuming the Data subjects request from AWS snowflake tables and post the data to adobe analytics privacy API.
- Research and experiment with new AI techniques and frameworks to improve existing models.
- Design, develop, and deploy machine learning models for various business applications.
- Developed the python script to automate the data cataloging in Alation data catalog tool. Tagged all the Personally identified Information (PII) data in Alation enterprise data Catalog tool, to identify the sensitive consumer information.
- Tackled highly imbalanced Fraud dataset using sampling techniques like under-sampling and over-sampling with SMOTE using Python Scikit-learn.
- Experimented with ensemble methods to increase the accuracy of the training model with different bagging and boosting methods and deployed the model on AWS.
- Created and maintained reports to display the status and performance of deployed model and algorithm with Tableau.
- Collect, clean, and preprocess large datasets for training AI models.
- Created an integration to consume the HBO consumer subscription information posted to AWS SQS- simple queue services and loaded into Snowflake tables for data processing, stored the meta data information into Postgres tables.
- Optimize machine learning algorithms for accuracy, performance, and scalability.
- Worked on generating the reports to provide the warner media brands consumer information to data subjects through python automation jobs.
- Maintain documentation and reports for AI models, experiments, and workflows.
- Involved in different phases of data acquisition, data collection, data cleaning, model development, model validation, and visualization to deliver solutions.

Environment: Python, PostgreSQL, AI/ML, AWS (Snowflake, SQS), Alation Data Catalog, Adobe Analytics Privacy API, Scikit-learn, SMOTE, Tableau, Bagging & Boosting Techniques, Ensemble Methods, Snowsql

Client: First Tech Federal Credit Union, San Jose, CA

January 2021 – March 2023

Role: Data Scientist

Responsibilities:

- Identifying the facts & dimensions; grain of fact, aggregate tables for Dimensional Models.
- Dimensional Data Modeling to deliver Multi-Dimensional STAR schemas.
- Developed dimensions and fact tables for data marts like Monthly Summary, Inventory data marts with various Dimensions like Time, Services, Customers and policies.
- Utilize MLOps practices for continuous integration, testing, and deployment of ML models.
- Evaluate machine learning models using metrics like precision, recall, F1-score, and AUC-ROC.
- Implement deep learning architectures such as CNNs, RNNs, and Transformers for advanced tasks.
- Devised simple and complex SQL scripts to check and validate Dataflow in various applications.
- Involved in setting virtual environment in Linux, write Linux script for process automation.
- Used Hive for creation of ORC formatted tables and used ADF for data orchestration to Azure database. Data copied from ADLS to Azure SQL database using ADF pipelines invoked using PowerShell scripting□ Created/Managed Groups, Workbooks and Projects, Database Views, Data Sources and Data Connections□ Developed dashboards using Power BI as per requirements.
- Write research reports describing the experiment conducted, results, and findings and make strategic recommendations to technology, product, and senior management. Worked closely with regulatory delivery leads to ensure robustness in prop trading control frameworks using Hadoop, Python Jupyter Notebook, Hive and NoSql.

- Ensure fairness, bias mitigation, and explainability in AI models.
- Developed logistic regression models (using R programming and Python) to predict subscription response rate based on customer's variables like past transactions, response to prior mailings, promotions, demographics, interests and hobbies, etc.
- Develop APIs and integrate AI models into software applications.
- Devised PL/SQL/Pig/Hive statements - Stored Procedures, Functions, Triggers, Views and packages. Made use of Indexing, Aggregation and Materialized views to optimize query performance.
- Designed, developed and delivered / implemented data solutions to include: architecture design, prototyping of concepts to proof of concept, development of standards, design and development of test plans, code and module design, development and testing, data solution debugging, design and implementation of a solution that follows efficient design techniques and development that meets and exceeds the intent of the design of the data solution.
- Performed all necessary day-to-day GIT support for different projects, Responsible for design and maintenance of the GIT Repositories, and the access control strategies.
- Performed Logical & Physical Data Modeling - ER Modeling & Dimensional Modeling.
- Knowledge in Tableau Administration Tool for Configuration, adding users, managing licenses and data connections, scheduling tasks, embedding views by integrating with other platforms.

Environment: PL/SQL, Python, R Programming, AI/ML, MLOps, Hive, Pig, Hadoop, NoSQL, Jupyter Notebook, Azure Data Factory (ADF), Azure SQL Database, ADLS, Power BI, Tableau, Linux, Shell Scripting, Git, SQL (Stored Procedures, Functions, Triggers, Views, Indexing, Aggregation, Materialized Views).

Client: HUMANA INC, Louisville, KY

November 2018 - December 2020

Role: Machine Learning Engineer

Responsibilities:

- Used SSIS to create ETL packages to Validate, Extract, Transform and Load data into Data Warehouse and Data Mart.
- Collaborate with data engineers and operation team to implement ETL process, write and optimize SQL queries to perform data extraction to fit the analytical requirements.
- Generated percentile scores for capturing distance between recurring incidents and integrated with complaints effectiveness metrics dashboard in Tableau to provide visual insights to business users.
- Designed and implemented supervised and unsupervised machine learning.
- Maintained and developed complex SQL queries, stored procedures, views, functions and reports that meet customer requirements using Microsoft SQL Server
- Developed a novel encoding technique to convert text data as input for use in machine learning models.
- Work with business stakeholders to refine and respond to their ad hoc requests and improve their existing reporting and dashboards as necessary.
- Participate in feature engineering such as feature generating, PCA, feature normalization and label encoding with Scikit-learn pre-processing.
- Performed data integrity checks, data cleansing, exploratory analysis and feature engineer using python and data visualization packages such as Matplotlib, Seaborn.
- Developed predictive models using Decision Tree, Random Forest and Naïve Bayes.
- Deal with missing values by column dropping, row dropping and replacing the missing data with statistical results.
- Explore and analyses the customer specific features by using Matplotlib and seaborn. Extract structured data from MySQL databases, developing basic visualizations or analysing A/B test results.

Environment: SSIS, SQL Server, MySQL, Python, Scikit-learn, Matplotlib, Seaborn, Tableau, ETL, Data Warehousing, Feature Engineering (PCA, Normalization, Label Encoding), Supervised & Unsupervised Machine Learning, Decision Trees, Random Forest, Naïve Bayes, KNN, K-Means Clustering, Hierarchical Clustering, Ensemble Methods, A/B Testing, Data Cleansing & Integrity Checks.

Client: Macy's Duluth, GA

February 2017 - October 2018

Role: Data Analyst

Responsibilities:

- Developed Machine Learning algorithm to diagnose blood loss.
- Perform analyses such as regression analysis, logistic regression, discriminant analysis, cluster analysis using SAS programming.
- Developed predictive models using Python & R to predict customers churn and classification of customers.
- Improved runtime of existing SQL queries by 60% maintaining code architecture to ensure smoothness and completeness of operations.
- Worked on Shiny and R application showcasing machine learning for improving the forecast of business.
- Design and deploy rich Graphic visualizations with Drill Down and Drop down menu option and Parameterized using Tableau
- Organization wanted to analyze construction data in order to make informed decisions on inventory stocking, tracking construction equipment and assets, cost optimization in order to marginalise profit and loss, and risk mitigation.
- Wrote SQL queries for operations such as Joining (LEFT, RIGHT, INNER, OUTER), Sorting (ORDER BY), Searching (WHERE, HAVING) and using Window Functions (Row Number, Rank, Dense Rank)
- Manipulating and Cleaning data using missing value treatment in Pandas and performed standardization
- Implemented Classification using Supervised algorithms like Linear Regression, Logistic Regression, Decision trees, KNN, Naive Bayes
- Experienced in developing business reports by writing complex SQL queries using views, volatile tables.

Environment: Python, R, SAS, SQL Server, DBVisualizer, Tableau, Shiny, Pandas, Machine Learning (Linear Regression, Logistic Regression, Decision Trees, KNN, Naïve Bayes, Clustering), Feature Engineering, Data Cleaning & Preprocessing, Business Intelligence & Reporting, Data Visualization (Drill-Down, Parameterized Dashboards), SQL (Joins, Sorting, Window Functions, Views, Volatile Tables), MS Office Suite – Excel.

Client: Mylan Laboratories, Hyderabad, India

August 2013 - Oct 2016

Role: Data Analyst

Responsibilities:

- Worked as a Subject Matter Expert for Membership, NCQA, and Symmetry data marts which consist of member data, risk scores.
- Worked on fixing the defects and enhancements of ETL jobs when failed in a production environment using SSIS (SQL Server Integration Services).
- Performed data analysis by doing data completion, data trending, data profiling using complex SQL queries, and maintained reports for daily/weekly/monthly production jobs.
- Involved in user acceptance testing (UAT) by preparing test cases, before/after scenarios, and sending them to business users for UAT approvals.
- Experience in creating and maintaining the Master and Child Packages model in SSIS for easier maintenance and better performance of the ETL strategy.

Environment: SQL Server Integration Services (SSIS), MS Excel, HP Service Manager, ETL, Data Analysis, Data Profiling, Data Completion, Data Trending, SQL Queries, User Acceptance Testing (UAT), Master and Child Package Models.